

Advanced MSc Capstone Project Student Guidelines

Duration: 3.75 months ($\approx$ 15 weeks, full time: **40 hours/week**)
Total Effort: ~600 hours

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1. Purpose of the Capstone Project

Your capstone project is **not about producing a long document**. It is about **demonstrating understanding, engagement, and professional-level project work** over the duration of the project.

For successful completion, you must be able to:

- *Explain* your ideas, methods, code, and results.
- Show how your work *developed* over the project period.
- *Defend* everything you submit.

2. Final Output

- **Primary submission:** a 10 to 15-page *research paper* (of approx. 10,000 words)
 - You will use a relevant journal paper template (Springer, IEEE etc.) based on the domain of your project. Your supervisor will advise you about this.
- Reflects **15 weeks of work** for a **30 ECTS project**.
- Your supervisor may act as a **co-author** of your paper if you decide to submit it to a relevant conference or journal after the work has been completed.
- **Note:** You may submit a full thesis as an alternative *only if you wish to*, but **all other guidelines in this document will still apply**. The length of the document does not carry any advantage, and the focus remains on your research contributions and your overall understanding.

3. Continuous Engagement (20%)

You must work **consistently throughout the project**, not just at the end.

You are required to:

- Attend regular supervisor meetings (usually one per week, on average).
- Submit **pre- and post-meeting reflections** for each meeting
- Maintain **dated notes, drafts, and code updates**.
 - You should use a combination of MS One Drive and MS Word to store an online version-controlled document containing all of your weekly notes.
- Be able to **explain your progress at every meeting**.

Marks are lost if:

- You miss, or do not arrange, a suitable number of meetings.

- You show poor preparation or poor organisational skills.
- You cannot explain your own work.

4. Project Milestones (60%)

You must pass **all milestones** by demonstrating your overall understanding and showing the full developmental history of your project.

Project Definition Document

Students provide a 1-page *Project Definition Document (PDD)* to the Programme Director, submitted through Canvas, informally describing their proposed project. You will be provided with a template for this, and your supervisor can advise you on the content.

The PDD as such is not graded but *a mark penalty of 10 marks out of 100 applies* if no satisfactory Project Definition Document is received by the deadline of **February 28th**.

Milestone 1 - Literature Review Presentation (10%) – **June 20th**

- Present your initial literature review in person to your supervisor.
- Approximately **10 secondary papers** should be included; and *no more* than 15.
- You must explain:
 - What *each* paper does.
 - Why it is *relevant* to your project.
- All notes and drafts must be available for review.

Milestone 2 – Primary Research Presentation (10%) – **July 20th**

- Present early results, code, and analysis.
- You must explain:
 - How your methods work.
 - What your code does.
- Show **incremental development** over time (not just one large upload)

Milestone 3 – Final Research Paper (40%) - **August 20th**

- Summarises all work completed.
- Must match the actual work done.
- Along with the submission of the paper itself, you must provide evidence for:
 - *Draft history* – You will complete the paper draft on Overleaf.

- *Code history* – You will need version control with regular commits.
- *Evolution of ideas* – Your work must demonstrate progress over time.

A milestone will **not pass** if:

- You cannot explain the work.
- There is no believable development history available.

5. Interactive Oral Assessment (20%)

- These will be scheduled for early September.
- These replace the traditional viva.
- This involves a **20-minute professional discussion** about the project.
- In this discussion, you act as the **project expert**.
- You will be asked about:
 - Your paper.
 - Your code and methods.
 - Your cited literature.

Rote memorisation will not work; understanding **is** essential to perform well.

6. Important Considerations

6.1 Use of Artificial Intelligence

- AI tools are **not banned**.
- However:
 - You must understand *everything* that you submit.
 - You must be able to explain it verbally.
 - AI-generated content that you cannot defend (or understand) **will fail**.

If you cannot explain your writing, code, or results, it will be treated as **inauthentic work**, and you will be referred to the School's Academic Integrity advisor. Your case will be reviewed, and you may be referred to the University's Academic Integrity Office with a risk of expulsion from the programme.

6.2 References

- You should identify ~10 (and a maximum of 15) secondary research papers.
 - You should focus on the quality of these papers and build up a full understanding of these chosen works and why they are relevant to your project. You will need to justify their importance and relevance.
- You must:
 - Have read them in detail.
 - Fully understand them.
 - Be able to explain and discuss their content.

Quality of understanding matters more than quantity.

6.3 English Language Requirement

You must be able to:

- Communicate clearly in spoken and written English without relying on tools.
- Meet the expected **6.5 IELTS equivalent level** with all communication.

If you:

- ..have written work that does not match your spoken ability.
- ..cannot explain your own writing.

This will be flagged early, and you may be referred for extra support.

6.4 What you must always be able to show

At *any* point in the project, you should be able to provide:

- Detailed notes and drafts.
- Incremental GitHub commits.
- Evidence of full engagement and steady progress.
- Clear explanations of your decisions throughout.

If you cannot show this, **marks will be lost** and milestones may fail.

7. Key Message for Students

You are assessed on **understanding, authenticity, and professional engagement**, not the volume of your writing. Work steadily, document everything, and make sure you can always explain your work.